

Clustering Lecture 9

Saturday, 23 November 2024

1:04 PM

Q.1

Passage:

You are given data for four observations A , B , C , and D with two features x_1 and x_2 . The values for each observation are as follows:

Observation	x_1	x_2
A	1	1
B	2	1
C	4	3
D	5	4

Using the K-means clustering technique, divide the observations into $K = 2$ clusters. Assume observations A and B are used as the initial centroids:

- Initial centroid for Cluster 1 (C1): (1, 1)
- Initial centroid for Cluster 2 (C2): (2, 1)

The Euclidean distance formula to calculate the distance between two points (x_1, y_1) and (x_2, y_2) is:

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Question: Based on the setup given, which cluster will observation C initially be assigned to?

- A) Cluster 1
- ☒ B) Cluster 2
- C) Cannot be determined from the given information
- D) Observation C is assigned to any of the clusters

$$\begin{aligned} \text{dis}(A, C) &= \\ \sqrt{4 + 4} &= \sqrt{8} \end{aligned}$$

$$\begin{aligned} \text{dis}(B, C) &= \\ \sqrt{4 + 4} &= \sqrt{8} \end{aligned}$$

$$\text{dis}(\underline{C2}, C) < \text{dis}(C1, C)$$

Q.2

Question: After assigning each observation to the nearest cluster based on initial centroids, let the updated centroids for both clusters be (x_1, y_1) and (x_2, y_2) . Calculate $x_1 + y_1 + x_2 + y_2$

(Round your answer to two decimal places).

A $\rightarrow$ C1 ✓

B $\rightarrow$ C2

C $\rightarrow$ C2

D $\rightarrow$ C2

$$\text{dis}(C1, D) = \sqrt{25} \quad , \quad \text{dis}(C2, D) = \sqrt{18}$$

Update Centroids:-

$$C1 = \{A\} : (1, 1)$$

$$C2 = \{B, C, D\} : \left(\frac{11}{3}, \frac{8}{3}\right)$$

$$x_1 + x_2 + y_1 + y_2 = 2 + \frac{19}{3} = \frac{25}{3} \approx \boxed{8.333}$$

Q.3

Question: After one iteration of updating the centroids as shown in Question 2, what will be the new assignments of the observations to the clusters?

- A) A → Cluster 1, B → Cluster 1, C → Cluster 1, D → Cluster 1
- B) A → Cluster 1, B → Cluster 1, C → Cluster 1, D → Cluster 2
- ☒ C) A → Cluster 1, B → Cluster 1, C → Cluster 2, D → Cluster 2
- D) A → Cluster 1, B → Cluster 2, C → Cluster 2, D → Cluster 2

$$C_1 = \underline{(1, 1)}$$

$$C_2 = \overset{3.67, 2.67}{\underline{\left(\frac{11}{3}, \frac{8}{3}\right)}}$$

$$\begin{array}{l} A (1, 1) \\ B \underline{(2, 1)} \end{array}$$

$$\begin{array}{l} C (4, 3) \\ D (5, 4) \end{array}$$

$$A \rightarrow C_1$$

$$C \rightarrow C_2$$

$$B \rightarrow C_1$$

$$D \rightarrow C_2$$

Q.4

In the K-means clustering algorithm, suppose we have:

- n : the number of data points (observations)
- k : the number of clusters
- d : the number of dimensions (features) for each data point
- i : the number of iterations until convergence

Given these parameters, which of the following options best describes the time complexity of the K-means clustering algorithm?

Options:

1. $O(n \cdot d)$
2. $O(n \cdot k \cdot d)$
- ✓ 3. $O(n \cdot k \cdot d \cdot i)$
4. $O(n^2 \cdot k \cdot d)$

$$\begin{array}{l} \text{Cent} \left(\frac{d}{\quad} \right) \\ \text{point} \left(\frac{d}{\quad} \right) \end{array}$$

for i iterations:

$$O(nkd) \text{ ① - Compute } d(C, P) \\ + \quad \checkmark \text{ centroids \& points}$$

$$O(nk) \text{ ② - assign the closest centroid}$$

$$O(nkd) \text{ ③ - Recompute centroids}$$

Step ①: For one centroid, point $\rightarrow O(d)$

For k clusters, n points $\rightarrow O(n \cdot k)$ pairs

Time to calculate dist $\rightarrow O(\underline{n \cdot k \cdot d})$

Step ②: n points.

1 point $\rightarrow O(k)$ to find min dist.

n points $\rightarrow \underline{O(n \cdot k)}$

$$\left(\frac{d}{\quad} \right) \left(\frac{d}{\quad} \right) \dots \left(\quad \right)$$

Step ③: k clusters:

1 cluster :- n points. Centroid :- $O(n \cdot d)$

$$\frac{n}{k} \cdot k \cdot d$$

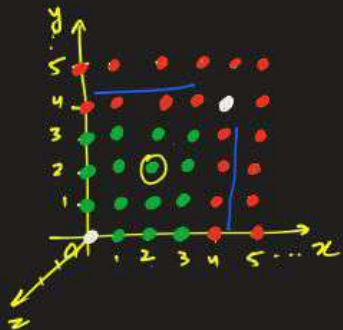
k clusters :- $O(n \cdot k \cdot d)$

$$\text{Overall time} = \boxed{O(i \cdot n \cdot k \cdot d)}$$

Q.5

A Chebyshev distance-based k-means clustering algorithm was applied to a dataset $D = \{(x, y, z) \mid x, y, z \in \mathbb{Z}, 0 \leq x, y, z \leq 9\}$ with $k = 3$.

If the points $(0, 0, 0)$ and $(4, 4, 4)$ are both part of Cluster 2, the **minimum number of points** that must be a part of Cluster 2 are _____



$$\text{Chebyshev's distance} = \max(|x_1 - x_2|, |y_1 - y_2|, |z_1 - z_2|)$$

$$\begin{aligned} \# \text{ points} &= 4 \times 4 \times 4 + 1 \quad (0 \rightarrow 3, 0 \rightarrow 3, 0 \rightarrow 3) + (4, 4, 4) \\ &= \boxed{65} \end{aligned}$$

Q.6

Passage:

Given the following dataset of points in a 2D plane:

Points: $P_1 = (1, 1)$, $P_2 = (2, 2)$, $P_3 = (3, 3)$, $P_4 = (6, 6)$, $P_5 = (8, 8)$, $P_6 = (10, 10)$.

The distance metric used is the Manhattan distance. Assume $k = 2$.

Question: Which of the following statements about K-Medoids is/are correct?

- ✓ A. K-Medoids is more robust to outliers compared to K-Means.
- ✗ B. The final medoids are guaranteed to be globally optimal.
- ✓ C. K-Medoids uses actual data points as medoids, while K-Means uses centroids that may not be actual data points.
- ✗ D. K-Medoids with Manhattan distance minimizes the squared sum of distances to medoids.



absolute

$$|x_1 - x_2| + |y_1 - y_2| + \dots$$

Q7

Question: If the initial medoids are $M1 = P1$ and $M2 = P4$. What is the assignment of points to clusters in the first round based on initial medoids?

- A) $P2 - M1, P3 - M1, P5 - M1, P6 - M2$
 B) $P2 - M1, P3 - M1, P5 - M2, P6 - M2$
 C) $P2 - M1, P3 - M2, P5 - M2, P6 - M2$
 D) $P2 - M2, P3 - M2, P5 - M2, P6 - M2$

$$\text{Cost} = 0 + 2 + 4 + 0 + 4 + 8$$

$$\text{old} = 18$$

	$M1 (1,1)$	$M2 (6,6)$
$(1,1) P1$	0	10
$(2,2) P2$	2	8
$(3,3) P3$	4	6
$(6,6) P4$	10	0
$(8,8) P5$	14	4
$(10,10) P6$	18	8

Q8

Question: If the initial medoids are $M1 = P1$ and $M2 = P4$. What is the percentage reduction in cost after the first update using k-Medoids algorithm?

(Round your answer to 2 decimal places)

$$M1: \{P1, P2, P3\},$$

$$M2: \{P4, P5, P6\}$$

$$P1: 0 + 2 + 4 = 6$$

$$P4: 0 + 4 + 8 = 12$$

$$P2: 2 + 0 + 2 = 4$$

$$P5: 4 + 0 + 4 = 8$$

$$P3: 4 + 2 + 0 = 6$$

$$P6: 8 + 4 + 0 = 12$$

$$\text{Cost}_{\text{new}} = 2 + 0 + 2 + 4 + 0 + 4 = 12$$

$$\% \text{ reduction} = (18 - 12) / 18 \times 100\% = \frac{1}{3} \times 100\% = 33.33\%$$

Q.9.

Consider a hierarchical clustering scenario using Single Linkage, Complete Linkage, and Average Linkage methods. A dataset consists of four points in a 2D plane:

$$P_1 = (1, 1), P_2 = (1, 4), P_3 = (5, 1), P_4 = (5, 4).$$

The distance metric used is **Euclidean Distance**. Initially, each point is treated as its own cluster.

Which of the following statements are correct regarding the clustering process?

- ☒ A. In the first iteration, using Single Linkage, P_1 and P_3 will form a cluster.
- ☒ B. Using Complete Linkage, P_1 and P_2 will merge before P_1 and P_3 .
- ☒ C. The distance between P_1 and the cluster $\{P_3, P_4\}$, when calculated using Average Linkage, is greater than 5.
- ☒ D. After the first iteration in Single Linkage, the minimum distance between clusters is 3.

	P_1	P_2	P_3	P_4
P_1	0	(3)	4	5
P_2	3	0	5	4
P_3	4	5	0	(3)
P_4	5	4	3	0



$$\text{avg dis}(P_1, \{P_3, P_4\}) = \frac{4+5}{2} = 4.5$$

Q. 10

Consider the following dataset of 5 points in a 2D space:

$$P_1 = (2, 3), P_2 = (4, 6), P_3 = (8, 9), P_4 = (10, 12), P_5 = (12, 15).$$

The distance metric used is **Euclidean Distance**. For the given data, the clustering starts with each point as its own cluster. The linkage criterion used is **Average Linkage**.

In the first iteration of clustering, calculate the distance between all pairs of clusters and determine the two clusters that will be merged first. What is the distance between these two clusters?

(Round to two decimal places.)

$$d(P_1, P_2) = \sqrt{13}$$

$$d(P_1, P_3) = \sqrt{72}$$

$$d(P_1, P_4) = \sqrt{145}$$

$$d(P_1, P_5) = \sqrt{244}$$

$$d(P_2, P_3) = \sqrt{25}$$

$$d(P_2, P_4) = \sqrt{72}$$

$$d(P_2, P_5) = \sqrt{145}$$

$$d(P_3, P_4) = \sqrt{13}$$

$$d(P_3, P_5) = \sqrt{52}$$

$$d(P_4, P_5) = \sqrt{13}$$

$$\text{distance} = \sqrt{13} = \boxed{3.6055}$$

Q.11

Consider the following statements about different linkage criteria used in hierarchical clustering. Identify all the correct statements.

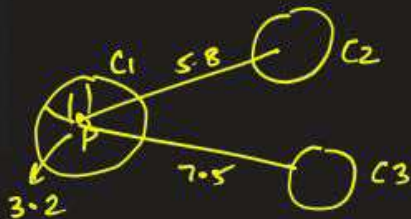
Statements:

- ☒ A. Single Linkage is prone to forming elongated clusters because it merges clusters based on the closest pair of points, regardless of the cluster shape.
- ☒ B. Complete Linkage tends to form compact and spherical clusters, as it merges clusters based on the maximum distance between points in the clusters.
- ☒ C. Average Linkage always results in clusters of equal size since it uses the mean distance between all pairs of points in the clusters.
- ☒ D. Single Linkage can produce the same result as Average Linkage if the data points are equidistant from one another.

Q.12

Consider a dataset with three clusters C_1 , C_2 , and C_3 . For a point $p \in C_1$, the following values are given:

- $a(p)$: Average intra-cluster distance within $C_1 = 3.2$
- $b(p, C_2)$: Average distance of p to points in cluster $C_2 = 5.8$
- $b(p, C_3)$: Average distance of p to points in cluster $C_3 = 7.5$



What is the Silhouette score for p ?

(Round your answer to two decimal places.)

$$\text{Silhouette Score} = \frac{5.8 - 3.2}{5.8} = \frac{2.6}{5.8} = \boxed{0.44827}$$

$$a(p) = 3.2$$

$$b(p) = 5.8$$

Q.13

Consider a dataset of 12 points divided into two ground truth classes:

- $L_1 = \{A, B, C, D, E, F\}$,
- $L_2 = \{G, H, I, J, K, L\}$.

A clustering algorithm produces two clusters:

- $C_1 = \{A, B, C, G, H\}$,
- $C_2 = \{D, E, F, I, J, K, L\}$.

$$\begin{aligned} |L_1 \cap C_1| &= 3 & |L_2 \cap C_1| &= 2 \\ |L_1 \cap C_2| &= 3 & |L_2 \cap C_2| &= 4 \end{aligned}$$

Calculate the **Purity** of the clustering result.

(Round to two decimal places.)

$$\text{Purity} = \frac{3+4}{12} = \frac{7}{12} = \boxed{0.5833}$$

Q.14

Consider a dataset of 12 points divided into two ground truth classes:

- $L_1 = \{A, B, C, D, E, F\}$,
- $L_2 = \{G, H, I, J, K, L\}$.

A clustering algorithm produces two clusters:

- $C_1 = \{A, B, C, G, H\}$,
- $C_2 = \{D, E, F, I, J, K, L\}$.

		Predicted	
		C_1	C_2
actual	L_1	3	3
	L_2	2	4

$$\frac{\# \text{ agreements}}{\text{Total pairs}} = \frac{TP + TN}{\binom{12}{2}}$$

Using the clustering and ground truth, calculate the **Rand Index (RI)**.

(Round to two decimal places.)

$$\begin{aligned} \text{Rand index} &= \frac{18+13}{66} = \frac{31}{66} \\ &\approx \boxed{0.4697} \end{aligned}$$

$$\begin{aligned} TP &= {}^3C_2 + {}^3C_2 + {}^2C_2 + {}^4C_2 \\ &= 3+3+1+6 = 13 \end{aligned}$$

$$FP = 3 \times 2 + 3 \times 4 = 18$$

$$FN = 3 \times 3 + 2 \times 4 = 17$$

$$\begin{aligned} TN &= {}^{12}C_2 - (13+18+17) \\ &= 18 \end{aligned}$$